

Algebra II with Trigonometry

Chapter 6.5

Mental Math (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4533652_alg-2trig-h-chp-6-5-note.pdf](#)

Domain of Inverse Sine 1. What is the domain of the inverse sine function?

Range of Inverse Cosine 2. What is the range of the inverse cosine function?

Range of Inverse Tangent 3. Does the range of the inverse tangent function include its endpoints, $-\pi/2$ and $\pi/2$?

Exact Value: Inverse Tangent 4. What is the exact value of $\tan^{-1}(1)$?

Exact Value: Inverse Sine 5. What is the exact value of $\sin^{-1}(-1)$?

Exact Value: Inverse Cosine 6. What is the exact value of $\cos^{-1}(-\frac{\sqrt{3}}{2})$?

Basic Cancellation Property 7. What is the exact value of $\sin(\sin^{-1}(\frac{1}{4}))$?

Domain Check / Undefined Expression 8. Is the expression $\sin(\sin^{-1}(5))$ defined or undefined?

Restricted Cancellation: Cosine 9. Keeping the range of inverse cosine in mind, what is the exact value of $\cos^{-1}(\cos(\frac{7\pi}{6}))$?

Restricted Cancellation: Tangent 10. Keeping the range of inverse tangent in mind, what is the exact value of $\tan^{-1}(\tan(\frac{2\pi}{3}))$?